

# **explainable AI Framework for Employee Attrition Prediction**

## **Project Team:**

Mahith 2420090095

Ch. Charan Teja 2420090022

G. Nishanth 2420030679

Moksha 2400090182

**Team :10**

**Course code : 24ALT3101**

**Course Coordinator : Dr. K. Swanthana**

## **Problem Statement**

- >Employee attrition is a major challenge for organizations because it increases recruitment costs, reduces productivity and creates difficulties in retaining skilled employees.
- > Traditional machine learning systems generally predict whether an employee is likely to leave, but often provide limited transparency about the reason behind the prediction.
- >HR managers need a system that not only identifies employees at risk, but also explains the factors influencing that risk.
- >Our project proposes an Explainable AI framework to make attrition predictions transparent and actionable.

## **Objectives**

- ◆ To Develop a Framework, utilizing machine learning model to predict whether an employee is likely to leave the organization.
- ◆ To improve model generalizability by evaluating the framework using appropriate datasets and validation approaches.
- ◆ To integrate Explainable AI techniques to provide transparent explanations for individual employee attrition predictions.
- ◆ To identify the key factors contributing to employee attrition and provide actionable insights for reducing employee turnover.
- ◆ To enhance prediction reliability by addressing class imbalance and evaluating model performance using suitable metrics.

Literature Survey

| Research Paper                                                                                                    | Techniques Used                                                                                                         | Purpose                                                                                                                               | Limitation                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| IBM Employee Attrition Analysis — Yang & Islam (2020) (arXiv)                                                     | Correlation Matrix, Random Forest, K-Means Clustering, Logistic Regression                                              | To identify the major factors influencing employee attrition and analyze why employees leave an organization.                         | Uses a single IBM dataset, so the findings may have limited generalizability to other organizations.                  |
| Predict Employee Attrition Using Multiple Machine Learning Techniques — Ghanem & Al-Alawi (2024) (DOI)            | Random Forest, XGBoost, Decision Tree, Bernoulli Naive Bayes, KNN, SVM, Logistic Regression, Oversampling, GridSearchCV | To compare multiple ML algorithms and identify the most effective model for employee attrition prediction.                            | Dataset has class imbalance and the results may not generalize well to other organizations; more datasets are needed. |
| Employee Attrition Prediction using Machine Learning: A Data-Driven Approach for Workforce Analytics (2025) (DOI) | Logistic Regression, Gradient Boosting, Random Forest, SVM, Gaussian Naive Bayes, Feature Engineering                   | To predict employee attrition and identify important employee characteristics that can help HR managers develop retention strategies. | The study is primarily focused on prediction and model comparison; broader real-world validation is still needed.     |
| A Stacking Ensemble Framework for Predicting Employee Turnover: Explainable AI with                               | Machine Learning, Ensemble Learning, Deep Learning, Stacking, SHAP                                                      | To predict employee turnover using ensemble approaches and explain the model's predictions using SHAP.                                | Relies on the IBM HR Analytics dataset, which limits generalizability to different organizations and                  |

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| SHAP — Rahman et al. (2025/2026 publication record) (EurekaMag)                                                |                                                                                        |                                                                                                                         | workforce environments.                                                                                                                          |
| Integrating Machine Learning and Explainable AI for Employee Attrition Prediction in HR Analytics (2026) (DOI) | Machine Learning, SHAP, Feature Selection, Data Balancing, Hyperparameter Optimization | To combine accurate attrition prediction with explainable AI so HR teams can understand the factors behind predictions. | Synthetic datasets such as IBM HR Analytics may not represent real-world organizations; SHAP may not fully capture complex feature interactions. |

| Research Paper                                                                                                           | Techniques Used                                                                         | Purpose                                                                                            | Limitation                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Employee Attrition Prediction Using Bayesian Optimized Stacked Ensemble Learning and Explainable AI — Konar et al., 2025 | Stacking, XGBoost, Random Forest, Bayesian Optimization, SHAP, LIME                     | To improve attrition prediction and explain why employees are predicted to leave.                  | Mainly evaluated on benchmark data; broader organizational validation is still needed. (Springer)                     |
| Integrating Machine Learning and Explainable AI for Employee Attrition Prediction in HR Analytics — Al-Ali et al., 2026  | Adaptive Boosting, Histogram Gradient Boosting, SHAP, feature selection, data balancing | To combine accurate attrition prediction with explainability, fairness and actionable HR insights. | Existing HR datasets may not fully represent different real-world organizations. (PubMed Central (PMC))               |
| Explainable AI for Employee Turnover Prediction: A SHAP-Based Intelligent Analytics Approach — Haojie Jia, 2026          | XGBoost, Random Forest, LightGBM, Logistic Regression, SHAP, SMOTE, fairness audit      | To provide global, local and interaction-based explanations for employee turnover predictions.     | Feature importance and subgroup effects can vary between datasets, requiring validation for each deployment. (Nature) |
| Predicting Employee Attrition:                                                                                           | Transformer, GAN-based                                                                  | To handle class imbalance and                                                                      | Authors identify the need for validation                                                                              |

|                                                                                                       |                                                            |                                                                                                             |                                                                                            |
|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| XAI-Powered Models for Managerial Decision-Making — 2025                                              | oversampling, SHAP                                         | provide transparent attrition predictions for HR managers.                                                  | across more industries and lightweight real-time implementation. (MDPI)                    |
| Employee Attrition Prediction: An Explanatory and Statistically Robust Ensemble Learning Model — 2026 | SHAP, LIME, Optuna, Random Oversampling, Ensemble Learning | To combine accurate prediction with global and local explanations and identify important retention factors. | Uses benchmark datasets, so real-world organizational validation remains important. (MDPI) |

## Research Gap Identification

- > **Limited Generalizability:** Most studies rely heavily on the IBM HR Analytics dataset, making it difficult to confirm whether the models perform effectively across different organizations and workforce environments.
- > **Limited Explainability:** Several existing approaches focus primarily on prediction accuracy and model comparison, without providing sufficient explanations for individual employee predictions.
- > **Lack of Actionable Insights:** Existing models mainly identify employees who are likely to leave but provide limited guidance on which factors should be addressed to reduce attrition.
- > **Class Imbalance Challenge:** Employee attrition datasets often contain significantly fewer employees who leave compared with those who stay, which can affect prediction performance.
- > **Limited Real-World Validation:** Many approaches are evaluated on benchmark or synthetic datasets rather than validated using diverse real-world organizational data.

## **Innovation**

- ◆ Integrates machine learning-based attrition prediction with Explainable AI.
- ◆ Provides understandable explanations for individual employee risk predictions.
- ◆ Identifies important employee attributes contributing to the predicted outcome.
- ◆ Supports aggregate analysis to identify departments with high attrition rates.
- ◆ Can reveal recurring causes, workforce trends and areas requiring management attention.

## **Creativity & Novelty**

- ◆ Creativity: Converts complex machine learning decisions into information that can be understood by HR personnel.
- ◆ Novelty: Combines individual prediction, prediction confidence and explainability within one framework.
- ◆ Uses multiple possible explanation approaches such as SHAP, LIME, feature importance and permutation importance.
- ◆ Extends the analysis from individual employees to department-level workforce patterns.
- ◆ Focuses on producing actionable insights rather than only a binary prediction.

## **Feasibility**

- ◆ Technical feasibility: The project uses established supervised machine learning and Explainable AI techniques.
- ◆ Data feasibility: The IBM HR Analytics Employee Attrition dataset is identified as a benchmark dataset in the project.
- ◆ Implementation feasibility: Data preprocessing, model training, evaluation and visualization can be implemented using Python-based tools.
- ◆ Evaluation feasibility: Accuracy, precision, recall, F1-score, ROC-AUC and confusion matrix can be used for model evaluation.
- ◆ Project scope is suitable for an academic implementation and can be extended with a dashboard or interface.

## **Project Plan**

- ◆ Phase 1 — Literature survey & requirements
- ◆ Phase 2 — Dataset collection & preprocessing
- ◆ Phase 3 — EDA & feature engineering
- ◆ Phase 4 — ML model development
- ◆ Phase 5 — Explainable AI integration
- ◆ Phase 6 — Model evaluation
- ◆ Phase 7 — Interface/dashboard development
- ◆ Phase 8 — Testing, documentation & presentation